

Report 240

All Order Asymptotics for Hypergraphs without Singleton Intersections

Exceptional components, two fluctuation scales, and inversion

5 October 2026

Abstract

Let A_n count simple 3-uniform hypergraphs on the labelled set $[n]$ in which distinct edges never meet in one vertex, and let B_n count those with no isolated vertices. These are OEIS A323297 and A323296. Starting from Howroyd’s component decomposition, we give an explicit finite algorithm for every fixed saddle correction, with relative remainder $O_J((r/n)^{J+1})$ at the exact saddle. We reverse the counting scale through three inverse-logarithmic corrections and give an arbitrary-order recursion, then distinguish this coarse expansion from a much finer exact-saddle inverse and rounding-safe two-ceiling bounds. For the vector of isolated vertices and the two nonsunflower tetrahedral types, we prove a globally summable conditional-likelihood expansion and the sharp product-Poisson total-variation equivalent

$$d_{\text{TV}} \sim \frac{10}{3} \frac{e^{-1/2} r^3}{\sqrt{2\pi} n}.$$

Separate saddle-shift estimates give relative probabilities of rare component-deletion events. Projecting away total vertex count resolves the nearly singular edge/component covariance: the component count has scale $\sqrt{n}/r$, while its edge-defect residual has the independent scale $r^2/\sqrt{3}$. A quantitative mixed Gaussian–Poisson test bound retains the entire exceptional vector without normalizing it. All results transfer to B_n at its own saddle. Complete coefficient expansions and conditioned Poisson representations are classical frameworks; no priority claim, effective numerical onset, or discrete-to-continuous total-variation claim is made. Exact bounded checks, noncertified numerical diagnostics, and a reproducible source archive accompany the proofs.

[write] **Status and trust boundary, added 7 October 2026, batch 113.** This is bundle Report 240 of an external AI-assisted research session, filed in ProveIt as the single-source research report `a323297-singleton-free-hypergraphs` (provenance, the sources as the write read them, what was checked, relation to the repository, the collected non-claims and the reading conventions are in Section 1.3). It is unrefereed and not formalized. Its title page and PDF author field read “Report 240”: it names no person or tool. *What rests on what.* The component classification (Proposition 1.1) is proved in the text and agrees with Howroyd’s generating functions in the OEIS; the conditioned Poisson representation and the likelihood reduction are Arratia–Tavaré’s, correctly applied. Every asymptotic statement has a conventional proof: full-arc Fourier domination, saddle-point expansion with integrated Taylor remainders, local limit estimates, saddle shifts, Schur-complement projection, and the mean value theorem. No proof uses a computation; the exact checks corroborate finite identities and the floating diagnostics are illustrations. The write adds Remarks 4.3 and 10.1 and dated notes; no statement, proof or number of the source was changed.

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1 The model and the classical component input

An admissible hypergraph is a set of distinct 3-subsets of $[n]$ such that any two distinct hyperedges have intersection of size 0 or 2. Labels are distinguished; no quotient by vertex permutations is taken. Isolated vertices are allowed in A_n and forbidden in B_n , with $A_0 = B_0 = 1$. We use the uniform probability measure on the A_n objects and write $\mathbb{P}_n, \mathbb{E}_n$ for its probability and expectation.

Andrew Howroyd's comments of 18 August 2019 in OEIS A323296 and A323297 give the component decomposition and the exponential generating functions used here [2, 1]. The connected count is the related sequence A323294, whose eventual formula is credited to Gus Wiseman [3]. We include the short classification to fix every small-size multiplicity.

Proposition 1.1 (Component classification). *Every nontrivial connected component is a single edge, a pair-star consisting of all triples formed by adjoining one of its other vertices to a common pair, or a three- or four-edge hypergraph on four vertices. On a fixed labelled m -set the connected counts are*

$$c_1 = 1, \quad c_2 = 0, \quad c_3 = 1, \quad c_4 = 11, \quad c_m = \binom{m}{2} \quad (m \geq 5).$$

The eleven four-vertex components comprise six two-edge pair-stars, four three-edge tetrahedra, and one complete tetrahedron.

Proof. Use adjacency of edges by nonempty intersection. If two edges each meet a third in two vertices, they share at least one vertex, so admissibility forces their own intersection to have size two. Thus adjacency is transitive, and every connected edge component is pairwise two-intersecting.

Choose edges abc, abd . An edge meeting both in two vertices is either abx , or acd , or bcd . If a vertex x outside $\{a, b, c, d\}$ occurs, the edge abx rules out acd and bcd , forcing all edges to contain ab . Every vertex of the component outside this pair then gives exactly its triple with the pair. Otherwise the component is confined to four vertices. A single edge has no uniquely distinguished pair and must be counted only once. For a pair-star on at least four vertices the pair is unique, giving $\binom{m}{2}$ choices. On four vertices the additional choices are the four triples of edges and the one complete edge set. $\square$

Let $\mathcal{D} = z \, d/dz$ and set

$$C(z) = \sum_{m \geq 1} c_m \frac{z^m}{m!} = z - \frac{z^2}{2} - \frac{z^3}{3} + \frac{5z^4}{24} + \frac{z^2 e^z}{2}, \quad (1.1)$$

$$A_n = n! [z^n] e^{C(z)}, \quad B_n = n! [z^n] e^{C(z) - z}. \quad (1.2)$$

Although the last expression for C includes negative polynomial terms, its power-series coefficients are nonnegative. This distinction matters when using characteristic-function domination. Write

$$H(z) = \frac{z^2 e^z}{2}, \quad U(z) = z - \frac{z^2}{2} - \frac{z^3}{3} + \frac{5z^4}{24}.$$

The exact positive saddle $r = r_n$ is defined by

$$\mathcal{D}C(r) = n, \quad \kappa_j = \mathcal{D}^j C(r), \quad b = \kappa_2, \quad \varepsilon = \frac{r}{n}. \quad (1.3)$$

It exists uniquely for $n > 0$, since $\mathcal{D}C$ is strictly increasing from zero to infinity. Throughout, limits are as $n \rightarrow \infty$; constants in fixed-order estimates may depend on that order or on the fixed polynomial under discussion.

1.1 The statistics and their exact defect identity

Let K be the number of components, including isolates, and E the number of hyperedges. Let S, T_3, T_4 count, respectively, isolated vertices, three-edge tetrahedral components, and complete four-edge tetrahedral components. Put $X = (S, T_3, T_4)$. The components counted by T_3, T_4 will be called nonsunflower components: all other nontrivial components have a common pair, with the single-edge case interpreted separately.

The contribution of a component with m vertices and e edges to $E + 2K - n$ is $e + 2 - m$. It is zero on every single edge and pair-star, one on an isolate or a three-edge tetrahedron, and two on a complete tetrahedron. Hence the exact identity is

$$\Delta := E + 2K - n = S + T_3 + 2T_4. \quad (1.4)$$

For example, the connected EGF marked by its number of edges is

$$z + \frac{vz^3}{6} + \frac{z^2}{2}(e^{vz} - 1 - vz) + \frac{(4v^3 + v^4)z^4}{24}.$$

Multiplying this connected series by a component marker and exponentiating yields the joint edge/component EGF. The simpler identity (1.4) explains why treating the raw pair (K, E) as a nondegenerate bivariate Gaussian would miss an entire fluctuation scale.

1.2 Scope and attribution

Hayman’s method and its complete-expansion refinements, notably Harris–Schoenfeld and Odlyzko–Richmond, supply the established analytic setting [4]. Refined independent Poisson conditioning and exact total-variation likelihood identities are likewise prior tools of Arratia–Tavaré [5]. The results below specialize these frameworks with explicit coefficients and remainders, a sharp exceptional-vector constant, rare-deletion corrections, and projected marked laws. The forbidden-intersection extremal literature studies a different question, the maximum number of edges [6]. A bounded source review did not identify this exact package of refinements, but no exhaustive novelty or priority assertion is intended.

1.3 Provenance, sources and reading conventions

[write] **Added 7 October 2026, batch 113.** *Provenance.* This report is Report 240 of the session bundle that ProveIt’s docs/incoming intake processed as batch 113: the archive `Report240.zip` (601,277 bytes, 30 files, wrapper directory `Report240/`), arrival commit `60f54ea06`, placed in `a4186a946`. The text is the delivered `article.tex` (52 lines) with its twelve `\input` files under `sections/` (21 pages, dated 5 October 2026), shipped under the same names. The programs are shipped under `code/` (the delivered root `build.py` as `code/build.py`); the four receipts, delivered in `code/`, and the requirements under `data/`; `COMPUTATION.md` and `SOURCES.md` as delivered. Not shipped, and retrievable from the arrival commit: the delivered `Report240.pdf`, the checksum manifest `MANIFEST.sha256` (29 entries, all verified at the write) and the delivered `README.md`, which the report `README` replaces. The package records no ProveIt commit; it carries no “prepared for private review” line and no personal data. It is a single-source report, so no merge choice was made. The write prefixed the 83 delivered labels with `sfh:` and updated the references to them (71 `\eqref`, 15 `\ref`); it added this subsection, the notes marked [write] and Remarks 4.3 and 10.1, each the last statement of its section, and inserted no numbered display, so every section, theorem and equation of the source keeps its delivered number.

The sources, as the write read them (7 October 2026): OEIS A323297, A323296 and A323294 in the internal format, with their b-files (Remark 10.1); the transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex` (p0:def:model, p0:def:three-inverses, p0:thm:staircase, plt:def:lw-monomial-log-datum, plt:thm:lw-template). Not read by the write: Odlyzko–Richmond, Arratia–Tavaré and Keevash–Mubayi–Wilson; the source’s account of them (Section 1 and SOURCES.md) is reported, not confirmed.

What was checked at intake. The batch-113 dossier read the manuscript, found no demonstrably wrong claim and reran the delivered suite on a copy (it reproduces apart from Windows path tests). *What the write checked.* It read every proof and found no error. With its own code (Python 3.14.4, exact integers and fractions; SymPy 1.14.0; mpmath 1.3.0 at 30–50 digits), not importing the delivered programs, it

- computed A_n, B_n by (10.1) for $n \leq 1200$, equal to all 201 terms of the b-files of A323297 and A323296, and the connected counts c_m , equal to all 1001 terms of A323294 [independent check, 7 October 2026: equal for $2 \leq m \leq 1000$. A323294 counts spanning hypergraphs, so its $a(0) = 1$ and $a(1) = 0$ are its own conventions, while here $c_1 = 1$ is the isolated vertex. Its e.g.f. is $1 + C(x) - x$]; A_n again by the second route of Section 10 for $n \leq 60$; the series identity (1.1); and, by enumerating every admissible edge set for $n \leq 8$, A_n, B_n , the connected counts and the defect identity (1.4) on every object;
- derived (3.3), (3.4) and DE_1 from the contraction rule (3.1) symbolically, with the odd orders vanishing; the recurrence (2.2); the identity for V before (7.3); the expansion of g and, by its own series reversal of (4.10), both corrections in (4.11) and all three in (4.8);
- evaluated, at the exact saddle and for $n = 100$ to 1200 with exact A_n , the remainders of (3.2) for $J = 0, 1, 2$ divided by $(r/n)^{J+1}$ (bounded: at most 0.19, 0.0043, 0.024, decreasing), the deletion formulas (6.7) and (6.5) against exact B_n/A_n and exact counts without tetrahedral components (remainders times n^2/r^2 and n^2/r^{11} between 0.14 and 0.21, and between -0.08 and -0.05), and $\log A_n - n g(r) = O(r^5)$;
- recomputed the table of Section 10 from the exact conditioning formula (5.5) and exact moments of K (note there), and checked $\mathbb{E}|1 - Z^2| = 4\phi(1)$ by quadrature; the component sums behind (5.1), (5.2), (7.1) and the deletion coefficients of (6.5), (6.7) by hand.

It reran, on a fresh copy of the delivered layout, `exact_counts.py`, `symbolic_coefficients.py` and `diagnostics.py` (normal and -0): the receipts equal the shipped ones as JSON. The guard suite stops on Windows with “backslash output components are forbidden”, a path test, not a mathematical failure. The PDF and ZIP builder and `reproduce_zip.py` were not run. Floating computations are observations, not certificates.

Relation to the repository. Before batch 113 no file of the repository named A323297, A323296 or A323294, and no report of the collection treats intersection-restricted hypergraphs; no result is shared and no reciprocal note is proposed. No Lean or Rocq file treats these sequences; no statement of this report is formalized, and its place in the collection confers no formal status. The inverse is compared with the transseries volume in Remark 4.3.

[write] What is not claimed, collected from the source. The component classification and the generating functions are prior (Howroyd, 2019; Wiseman’s formula for the connected counts); Hayman’s method with the Harris–Schoenfeld and Odlyzko–Richmond refinements and the conditioned Poisson and likelihood tools of Arratia–Tavaré are prior frameworks; the extremal question of Keevash–Mubayi–Wilson is different. No exhaustive novelty or priority claim. Not claimed: convergence of the expansion as $J \rightarrow \infty$; numerical values of K_J or an onset; that rounding a computed inverse certifies a threshold; a canonical smooth interpolation of the sequence; relative accuracy from the coarse logarithmic inverse; total-variation convergence of a discrete count to a continuous law; growing deletions or growing Fourier parameters; any coupling across n or almost-sure statement; that the diagnostics are certified. *The write adds:* its checks are exact or floating computations; its transseries remark claims no novelty for any inversion.

Reading conventions (letters the source uses in more than one sense, with the tempting false readings; no

symbol was renamed).

Symbol	Meanings
A, B, b	A_n, B_n the counts; $A_1(G)$ the first local correction in Lemma 5.2; $b = \kappa_2$ and b_G, b_0, b_x its versions. <i>False reading:</i> $A_1(G)$ is not the count A_1 .
C, c	$C(z)$ the connected EGF, C_0, C_B its deletions; c_m connected counts; $c = \log 2$ in Section 4; c, c_0, C constants; $c_* = \frac{10}{3}\phi(1)$ in Section 10.
$E, \mathbb{E}$	E the number of hyperedges; E_ℓ the saddle corrections (3.1); $\mathbb{E}$ expectation. <i>False reading:</i> E_1 is a coefficient, not an edge count.
K, L, M	K the number of components, K_0 in the hybrid proof, and K_J the remainder constant of Theorem 4.1; $L = \log y$, $L_X = S + T_3 + T_4$; M the threshold in Theorem 4.1 and $M = \sum jk_j$ in (3.1).
N	N, N_G, N_0 Boltzmann total sizes; $N(M)$ the integer threshold.
P, Q	P_j the polynomials (2.2); $P(z) = z + 5z^4/24$ in Section 5 and a general deleted polynomial in Section 6 (the source says so); Q a polynomial perturbation in Section 2, the product Poisson law (5.1), and Q_* a projected sum. <i>False reading:</i> the law Q is not the polynomial Q .
R, r	$R(X), R_G$ remainders, R_m the shift ratios (8.5); r the saddle, r_x, r_W, r_B its versions.
S, T	S the isolates and S_J the partial sums; T_3, T_4 tetrahedral counts, T the Schur complement (7.2) and T_0 a parameter bound. <i>False reading:</i> T is not $T_3 + T_4$.
V, v, W	V the Schur variance (7.2), V_W , and V_j phase terms; $v = \text{Var}_Q W$ and v the edge marker in Section 1.1; W the exceptional mass (5.2).
d, δ, ρ	d the defect mean (7.1), the degree of P in Theorem 6.1, d_α component defects; $\delta = W - \mu$ and the target displacement in Lemma 5.2; ρ the correlation (7.2) and the shifted saddle in Section 6.
$\kappa_j, k_j, \varepsilon$	$\kappa_j = \mathcal{D}^j C(r)$; $k_j = \mathcal{D}^j G(r)$ for $G = C$ or C_0 ; $\varepsilon = r/n$.

[write] **Independent check of the write, 7 October 2026.** An independent adversarial check of the batch-113 write (8748cfc59) used its own code. It read A323297 (#14), A323296 (#15) and A323294 (#34) again. The quotations of Remark 10.1 are verbatim, and so is the write’s observation that the component comment is in A323296 while A323297 carries the e.g.f.

- *Counts.* The check enumerated every admissible edge set for $n \leq 7$, a separate backtracking code. It reproduces A_n, B_n, c_m and the defect identity (1.4) on every object. Power-series exponentiation of (1.1) reproduces all 201 b-file terms of A323297 and of A323296. For A323294 see the bracketed precision above.
- *Expansions.* At the exact saddle, with counts computed to 80 digits for $n = 100, \dots, 1200$, the remainders of (3.2) for $J = 0, 1, 2$, the deletion remainders of (6.7) and (6.5), and $(\log A_n - n g(r))/r^5$ all equal the write’s values.
- *Inverse.* An own symbolic reversion of (4.10) gives exactly the corrections of (4.11) and (4.8). Remark 4.3(2) was checked against `plt:def:lw-monomial-log-datum`: $\rho = 1, \mu = 4, \beta = -\log 2$, and the perturbation $\log(1 + 2t) + \log(g/r)$ lies in $t\mathbb{Q}(\log 2)[\log r][[t]]$.
- *Table of Section 10.* All sixteen entries were recomputed from the exact conditioning formula (5.5) and exact moments of K , and agree with the write’s values to eight digits. The five values named in the note are roundings and the other eleven are truncations. $\mathbb{E}|1 - Z^2| = 4\phi(1)$ was confirmed.

Also confirmed: the provenance (601,277 bytes, 30 files, 52 lines, 21 pages, 29 manifest entries); the byte identity of the staged files; the README listing; and the numbering, with 83 delivered labels and 0 differences against the `.aux` of a build of the placed text. One precision was added in bracketed notes, on the A323294 comparison, in this section and in Remark 10.1. No claim of the write was found wrong.

2 Saddle scales and control of every Fourier arc

The same estimates will be used for several connected EGFs. In this section let $G = H + Q$ for a fixed real polynomial Q , with nonnegative power-series coefficients, and suppose $a = \mathcal{D}G(r) \sim n$. This includes C , $C - z$, the exceptional-component deletion, and any other fixed deletion retaining nonnegative coefficients. Terms of polynomial degree larger than four change only harmless polynomial bounds below.

Lemma 2.1 (Cumulant scales). *For each fixed $j \geq 1$,*

$$\mathcal{D}^j G(r) \sim H(r)r^j \sim nr^{j-1}, \quad b_G := \mathcal{D}^2 G(r) \sim nr, \quad r \sim \log n. \quad (2.1)$$

Every fixed power of r is $o(n^a)$ for every $a > 0$.

Proof. Define polynomials P_j by

$$P_0(z) = 1, \quad P_{j+1}(z) = (z+2)P_j(z) + zP'_j(z). \quad (2.2)$$

Then $\mathcal{D}^j H = HP_j$, and P_j is monic of degree j . Fixed polynomial perturbations are negligible compared with H . In particular

$$a = \frac{e^r r^2 (r+2)}{2} + O(r^q)$$

for a fixed q . Since $a \sim n$, we have $\log a = \log n + o(1)$, and logarithms give $r + 3 \log r - \log 2 + o(1) = \log n$. This proves all assertions. $\square$

For the Boltzmann total size N_G , independent Poisson component counts have cumulant generating function $G(re^t) - G(r)$. Its centered characteristic function is

$$\chi_G(\theta) = \exp\{G(re^{i\theta}) - G(r) - i\theta \mathcal{D}G(r)\}.$$

Lemma 2.2 (Full-arc domination). *There are $c, c_0 > 0$, valid for all sufficiently large r , such that*

$$|\chi_G(\theta)| \leq e^{-cb_G \theta^2} \quad (|\theta| \leq c_0/r), \quad (2.3)$$

$$|\chi_G(\theta)| \leq e^{-cn/r^2} \quad (c_0/r \leq |\theta| \leq \pi). \quad (2.4)$$

Proof. Under the weights of H , component size has distribution $m = 2 + \text{Pois}(r)$. The sizes $m \in [r/2, 2r]$ carry $1 - o(1)$ of both its mass and its weighted second moment. To see the exponential tail estimate directly, apply Markov's inequality to $e^{t \text{Pois}(r)}$, whose expectation is $\exp(r(e^t - 1))$, with a suitable fixed positive or negative t . Multiplying by any fixed power of m does not change exponential decay.

Choose c_0 small enough that $|m\theta| \leq 1$ on this size window. The loss in the real part of the exponent is the sum of nonnegative terms $\lambda_m(1 - \cos m\theta)$. Keeping just this window and using $1 - \cos u \geq cu^2$ gives (2.3). The finitely many polynomial modifications lie outside the window for large r .

For the outer arc, regardless of the phase of $H(re^{i\theta})$,

$$\Re\{H(re^{i\theta}) - H(r)\} \leq -H(r)\{1 - e^{-r(1 - \cos \theta)}\}. \quad (2.5)$$

On $[-\pi, \pi]$, use $1 - \cos \theta \geq 2\theta^2/\pi^2$ and $1 - e^{-x} \geq c \min(x, 1)$. For $|\theta| \geq c_0/r$, the right side is at most $-cH(r)/r \asymp -n/r^2$. The polynomial change contributes $O(r^q)$, which is absorbed for large r . $\square$

Thus no secondary lattice arcs are being omitted. In particular, removal of sizes one or two does not invalidate the proof: every sufficiently large consecutive size is present. The point is stronger than a statement of aperiodicity; it supplies the exponentially small bound required after coefficient extraction and conditioning.

3 An explicit expansion to every fixed order

Write $\kappa_j = \mathcal{D}^j C(r)$ and $b = \kappa_2$ at the exact saddle (1.3). For $\ell \geq 0$ define

$$E_\ell(r) = \sum_{\substack{k_3, \dots, k_{2\ell+2} \geq 0 \\ \sum_{j=3}^{2\ell+2} (j-2)k_j = 2\ell}} (-1)^{M/2} (M-1)!! \prod_{j=3}^{2\ell+2} \frac{1}{k_j!} \left(\frac{\kappa_j}{j! b^{j/2}} \right)^{k_j}, \quad M = \sum_j j k_j. \quad (3.1)$$

The sum is finite, $M = 2\ell + 2 \sum k_j$ is even, and the convention $(-1)!! = 1$ gives $E_0 = 1$. Thus this is an executable finite coefficient rule, not an implicit existence statement. Put $S_J = \sum_{\ell=0}^J E_\ell$.

Theorem 3.1 (Complete fixed-order expansion). *For every fixed integer $J \geq 0$,*

$$A_n = \frac{n! e^{C(r)}}{r^n \sqrt{2\pi b}} \left\{ S_J(r) + O_J \left(\left(\frac{r}{n} \right)^{J+1} \right) \right\}. \quad (3.2)$$

Each $E_\ell = O_\ell((r/n)^\ell)$, so the error is relative as well as additive inside the braces. The first two corrections are

$$E_1 = \frac{\kappa_4}{8b^2} - \frac{5\kappa_3^2}{24b^3}, \quad (3.3)$$

$$E_2 = -\frac{\kappa_6}{48b^3} + \frac{7\kappa_3\kappa_5}{48b^4} + \frac{35\kappa_4^2}{384b^4} - \frac{35\kappa_3^2\kappa_4}{64b^5} + \frac{385\kappa_3^4}{1152b^6}. \quad (3.4)$$

Replacing C by $C - z$ and using its own exact saddle gives the same theorem for B_n .

Proof. Cauchy's formula on $|z| = r$ reduces the coefficient to the integral of

$$\exp\{C(re^{i\theta}) - C(r) - i n \theta\}.$$

Set $t = \theta\sqrt{b}$. By (2.1), its degree- j centered phase term, for $j \geq 3$, is

$$V_j(t) = \frac{i^j \kappa_j t^j}{j! b^{j/2}} = O_J(\varepsilon^{(j-2)/2} |t|^j).$$

Taylor expansion through degree $2J+3$ gives phase remainder $O_J(\varepsilon^{J+1} |t|^{2J+4})$. This bound follows from the all-real Taylor remainder for e^{iu} after summing the nonnegative component weights, so it does not require an unproved complex-neighborhood derivative estimate.

Assign weight $j-2$ to V_j . On $|t| \leq \varepsilon^{-\alpha}$, choose a sufficiently small fixed $\alpha > 0$ depending only on J , and expand $\exp(\sum V_j)$ through weighted degree $2J+1$. To justify the integrated remainder, first take the ordinary exponential Taylor expansion through total degree $2J+1$ and then discard terms of higher weighted degree. Every discarded finite monomial has weight at least $2J+2$. The remaining infinite Taylor remainder has the same factor ε^{J+1} times a fixed polynomial in $|t|$, because every phase term has positive weight and the sum of their absolute values is bounded on the chosen

window. The phase remainder is controlled the same way. Multiplication by $e^{-t^2/2}$ therefore gives an $L^1(dt)$ error $O_J(\varepsilon^{J+1})$.

Between this window and $c_0\sqrt{b}/r$, Lemma 2.2 gives a Gaussian tail smaller than every fixed power of ε . The outer arc has bound e^{-cn/r^2} , also negligible after its normalization. Gaussian-polynomial tails allow all retained integrals to be extended to the real line. Odd powers of t integrate to zero. A term of weight 2ℓ integrates to its standard Gaussian moment $(M-1)!!$ times the sign $i^M = (-1)^{M/2}$, giving (3.1).

The size estimate for each monomial is the product of the factors $O(\varepsilon^{(j-2)k_j/2})$, hence $O(\varepsilon^\ell)$. The same argument applies to $C-z$, whose fixed polynomial change preserves every Fourier estimate. $\square$

All κ_j needed in (3.1) are computed by the polynomial recurrence (2.2), followed by adding $\mathcal{D}^j U(r)$. There is no need to repeatedly differentiate exponential expressions symbolically. At each fixed order the expansion is finite, and its validity does not assert convergence as $J \rightarrow \infty$ with n fixed.

Remark 3.2 (Why the exact saddle is retained). The logarithmic approximation $r \sim \log n$ is adequate for describing scales, but replacing r by a coarse approximation inside $e^{C(r)}r^{-n}$ need not preserve relative count accuracy. Formula (3.2) always means the exact solution of (1.3). Likewise, the fact that deleting z is exponentially small compared with $H(z)$ does not make A_n and B_n asymptotically equal.

4 Logarithmic inversion and fine threshold location

Two distinct inverse problems must be separated. A reversed logarithmic expansion gives an explicit asymptotic size and a practical root-finding seed. The exact-saddle expansion gives a far finer location and, when supplemented with effective bounds or exact adjacent checks, integer thresholds.

4.1 The exact-saddle smooth function

For real x sufficiently large let r_x solve $\mathcal{D}C(r_x) = x$, put $b_x = \mathcal{D}^2C(r_x)$, and define

$$F_J(x) = \log \Gamma(x+1) + C(r_x) - x \log r_x - \frac{1}{2} \log(2\pi b_x) + \log S_J(r_x). \quad (4.1)$$

Since $S_J = 1 + O(r_x/x)$, it is positive eventually. Theorem 3.1 gives, at integer n ,

$$\log A_n = F_J(n) + O_J((r_n/n)^{J+1}). \quad (4.2)$$

Differentiating the saddle relation gives $dr_x/dx = r_x/b_x$. The implicit derivatives in $C(r_x) - x \log r_x$ cancel, and therefore

$$F'_J(x) = \psi(x+1) - \log r_x - \frac{\kappa_3(r_x)}{2b_x^2} + \frac{r_x}{b_x} \frac{d}{dx} \log S_J(r_x) \sim \log(x/r_x) > 0. \quad (4.3)$$

Indeed the explicit rational monomials in (3.1) satisfy $\mathcal{D}E_\ell = O_J(r\varepsilon^\ell)$, and the last contribution is $O_J(\varepsilon/x)$. Thus F_J has a unique inverse on a sufficiently large half-line.

Theorem 4.1 (Rounding-safe inverse brackets). *Fix J . Suppose K_J and a starting point are chosen so that*

$$|\log A_n - F_J(n)| \leq K_J(r_n/n)^{J+1}$$

for all integers beyond that point. For sufficiently large M , define x_-, x_+ by

$$F_J(x_-) + K_J(r_{x_-}/x_-)^{J+1} = \log M, \quad (4.4)$$

$$F_J(x_+) - K_J(r_{x_+}/x_+)^{J+1} = \log M. \quad (4.5)$$

Then the integer threshold $N(M) = \min\{n : A_n \geq M\}$ satisfies

$$\lceil x_- \rceil \leq N(M) \leq \lceil x_+ \rceil. \quad (4.6)$$

If x_J is defined by $F_J(x_J) = \log M$, the bracket width and each endpoint's distance from x_J have order at most

$$O_J \left(\frac{(r_{x_J}/x_J)^{J+1}}{\log(x_J/r_{x_J})} \right). \quad (4.7)$$

Proof. The two functions on the left of (4.4)–(4.5) are eventually increasing, with derivatives asymptotic to (4.3). If an integer $n < \lceil x_- \rceil$, then $n < x_-$, and the upper estimate for $\log A_n$ is below $\log M$. If $n \geq \lceil x_+ \rceil$, its lower estimate is at least $\log M$. The finite early segment is irrelevant for sufficiently large M . The mean value theorem and (4.3) give (4.7). Finally A_n is nondecreasing by adjoining an isolated labelled vertex, and the coefficient formula also implies eventual strict increase. $\square$

The theorem is an asymptotic bracket statement; this article does not supply numerical values for K_J and the starting point. It therefore does not claim that rounding a computed asymptotic inverse alone certifies an exact threshold. When the bracket crosses an integer, exact adjacent coefficient comparisons settle the answer. There is also no need to posit a canonical smooth interpolation of the integer sequence: (4.6) is the precise meaning of fine range location.

4.2 Three explicit inverse-logarithmic corrections

Let $y = \log M$, $L = \log y$, $\ell = \log L$, and $c = \log 2$.

Theorem 4.2 (Reversed logarithmic expansion). *For each fixed J , the solution $x_J(y)$ of $F_J(x) = y$ satisfies*

$$x_J(y) = \frac{y}{L} \left\{ 1 + \frac{2\ell + 1}{L} + \frac{4\ell^2 - 2\ell + c - 1}{L^2} + \frac{8\ell^3 - 22\ell^2 + 8c\ell - c - c^2/2 - 1/2}{L^3} + O\left(\frac{\ell^4}{L^4}\right) \right\}. \quad (4.8)$$

There is a coefficient-by-coefficient recursion for every fixed subsequent logarithmic order.

Proof. Define

$$\begin{aligned} n_0(r) &= \tfrac{1}{2}e^r r^2(r+2), \\ g(r) &= r + 2\log r - c - 1 + \log(1 + 2/r) + \frac{1}{r+2}. \end{aligned} \quad (4.9)$$

The exact saddle, Stirling's expansion, and the polynomial perturbation imply

$$n = n_0(r) + O(r^4), \quad F_J(n) = n_0(r)g(r) + O(r^5).$$

For example, the main factor in the latter expression is $\log n_0 - \log r - 1 + H(r)/n_0 = g(r)$; the saddle perturbation and fixed logarithmic factors fit inside $O(r^5)$. Errors polynomial in r are exponentially small relatively here. They may be discarded for fixed inverse-logarithmic orders, though not for count-relative accuracy. Taking logarithms and then dividing by g yields, to all such orders,

$$L = r + 3 \log r - c + \log(1 + 2/r) + \log g(r), \quad n = \frac{y}{g(r)}. \quad (4.10)$$

First,

$$g(r) = r + 2 \log r - c - 1 + \frac{3}{r} - \frac{4}{r^2} + \frac{20}{3r^3} + O(r^{-4}).$$

Reversion in (4.10) gives

$$\begin{aligned} r = L - 4\ell + c + \frac{14\ell - 3c - 1}{L} \\ + \frac{26\ell^2 - 12c\ell - 54\ell + 3c^2/2 + 12c + 7/2}{L^2} + O(\ell^3/L^3). \end{aligned} \quad (4.11)$$

Substituting (4.11) in $y/g(r)$ gives (4.8).

For a mechanical arbitrary-order rule set $t = 1/L$, regard ℓ, c as formal constants, and seek

$$R = t^{-1} - 4\ell + c + \sum_{k \geq 1} a_k(\ell) t^k, \quad \log R = \ell + \log(tR).$$

Expand (4.10) with $\log g(R) = \ell + \log(tg(R))$. At power t^k the new unknown a_k occurs with coefficient one in R , while its appearances in logarithms are delayed by at least one power. Solve successively, then expand $1/[tg(R)]$. This gives polynomials $q_k(\ell)$ of degree at most k in

$$n = \frac{y}{L} \sum_{k \geq 0} \frac{q_k(\ell)}{L^k}.$$

Taylor's theorem for $\log(1 + u)$, where $|u| = O(\ell/L)$, bounds a truncation through relative order m by $O_m(\ell^{m+1}/L^{m+1})$. The derivative of $\log(n_0(r)g(r))$ is $1 + O(1/r)$, so inversion preserves that order. This justifies the formal recursion as an asymptotic expansion. $\square$

The $\log 2$ terms record the factor $1/2$ in H . Even the error in (4.8) is eventually much larger than one. A leading Lambert- W seed $x \approx y/W(y)$ is still coarser, and neither seed should be substituted into a counting formula and called relatively accurate.

4.3 Newton refinement to the exact-saddle scale

Use (4.8) to initialize Newton iteration for the full function F_J , with derivative (4.3). On $|x - x_J| \leq x_J/2$, direct differentiation gives $F'_J \asymp L$ and $F''_J = O(1/x_J)$. Taylor's theorem therefore yields

$$|e_{k+1}| \leq \frac{C|e_k|^2}{x_J L}.$$

The logarithmic seed has vanishing relative error and eventually lies in this basin. Once the numerical residual is at most the coefficient-error scale $(r/x)^{J+1}$, the mean value theorem converts it to (4.7), subject to the numerical error margin. Solving the reduced equation $n_0 g = y$ alone would not obtain this finer accuracy. Effective integer certification still requires the additional bounds or exact checks specified above.

Remark 4.3 ([write] The inverse and the transseries volume, 7 October 2026). Compared statement by statement with the repository’s transseries volume [7]; the volume’s symbols are not used here.

1. *The growth.* $\log A_n = n g(r) + O(r^5)$ with $g(r) \sim r \sim \log n$ contains the term $n \log n$, so the sequences are not of the exponential–power form of `p0:def:model`; no instance of the volume’s model is claimed.
2. *The reversed logarithmic expansion.* Writing $\log g(r) = \log r + \log(g(r)/r)$ with $g(r)/r = 1 + (2 \log r - c - 1)/r + 3/r^2 - \dots$, the reduced equation (4.10) for r is literally the volume’s monomial–logarithmic equation (`plt:def:lw-monomial-log-datum`) in the unknown r : unit slope, logarithmic coefficient 4, constant $-\log 2$, and a perturbation in $1/r$ of positive order with coefficients polynomial in $\log r$. The three hypotheses of `plt:thm:lw-template` hold, so the expansion (4.11) of r (re-expanded from the template’s variable $L + \log 2$ into powers of $1/L$) is an exact formal instance of the template, and (4.8) follows from it by the explicit substitution $x = y/g(r)$. The write’s own reversion reproduces both. The reduction itself, valid to every inverse-logarithmic order, is the source’s argument in the proof of Theorem 4.2.
3. *The exact-saddle inverse.* F_J differs from the reduced model by terms polynomial in r , relatively exponentially small in r : below every order of the template’s chart. So x_J beyond (4.8), the Newton refinement and the brackets of Theorem 4.1 are not instances of `plt:thm:lw-template`; they are proved directly, by the mean value theorem for the explicit F_J .
4. *The thresholds.* $N(M)$ is the volume’s staircase (`p0:def:three-inverses(1)`) of the eventually increasing A_n . The functions $F_J \pm K_J(r_x/x)^{J+1}$ do not take the values $\log A_n$ at the integers, so they are not admissible interpolations, and Theorem 4.1, proved directly at the integers, is an analogue of `p0:thm:staircase(2)`, not an instance; the source’s refusal to posit a canonical smooth interpolation is the distinction of `p0:def:three-inverses`.

5 The exceptional vector and a sharp Poisson law

At the saddle r , let Q be the product law of independent Poisson variables with means

$$\lambda_1 = r, \quad \lambda_3 = \frac{r^4}{6}, \quad \lambda_4 = \frac{r^4}{24}. \quad (5.1)$$

Under this law define

$$\begin{aligned} W &= S + 4T_3 + 4T_4, & \mu &= \mathbb{E}_Q W = r + \frac{5r^4}{6}, \\ v &= \text{Var}_Q W = r + \frac{10r^4}{3}, & \delta &= W - \mu. \end{aligned} \quad (5.2)$$

We use $d_{\text{TV}}(P, Q) = \sup_A |P(A) - Q(A)| = \frac{1}{2} \int |dP/dQ - 1| dQ$.

Theorem 5.1 (Summable likelihood and sharp total variation). *The exact likelihood $h = d\text{Law}_n(X)/dQ$ satisfies*

$$h(X) = 1 + \frac{v - \delta^2}{2b} + \frac{\kappa_3 \delta}{2b^2} + R(X), \quad \mathbb{E}_Q |R(X)| = O(r^6/n^2). \quad (5.3)$$

Consequently, with $\phi(1) = e^{-1/2}/\sqrt{2\pi}$,

$$d_{\text{TV}}(\text{Law}_n(X), Q) \sim \phi(1) \frac{v}{b} \sim \frac{10}{3} \phi(1) \frac{r^3}{n}. \quad (5.4)$$

The comparison law has growing means. Small absolute total variation alone says nothing relatively accurate about an exponentially rare zero-count event. Those events are treated separately in Section 6.

5.1 Exact conditioning and the correct centering

The assembly construction gives independent Poisson counts of all component types with means equal to their connected EGF weights at r . Conditional on total vertex count $N = n$, they give exactly the uniform labelled assembly: multiplying the Poisson probabilities leaves the assembly factors after the common r^n and exponential normalization cancel. This is the standard refined conditioned-process representation [5, Sections 2–3 and 6].

Remove the three exceptional types by putting

$$P(z) = z + \frac{5z^4}{24}, \quad C_0(z) = C(z) - P(z), \quad N_0 = N - W.$$

The series C_0 has nonnegative coefficients and retains the single edge and all pair-stars. Since N_0 and X are independent,

$$h(X) = \frac{\mathbb{P}(N_0 = n - W)}{\mathbb{P}(N = n)}. \quad (5.5)$$

This exact likelihood reduction is also classical [5, Theorems 3 and 5]. The numerator is centered at its own mean $n - \mu$, so its target displacement is $-\delta$, not $-W$. That centering is responsible for the signed mass-zero quadratic term in (5.3).

5.2 A local expansion uniform over all targets

Lemma 5.2 (Global polynomial remainder). *For $G = C$ or C_0 at r , write $a_G = \mathcal{D}G(r)$, $b_G = \mathcal{D}^2G(r)$, and $k_j = \mathcal{D}^jG(r)$. Put*

$$A_1(G) = \frac{k_4}{8b_G^2} - \frac{5k_3^2}{24b_G^3}.$$

For every real δ such that $a_G - \delta$ is an integer, including a negative integer,

$$\sqrt{2\pi b_G} \mathbb{P}(N_G = a_G - \delta) = 1 + A_1(G) - \frac{\delta^2}{2b_G} + \frac{k_3\delta}{2b_G^2} + R_G(\delta), \quad (5.6)$$

where $x = \delta/\sqrt{b_G}$ satisfies the global bound

$$|R_G(\delta)| \leq C\{x^4 + \varepsilon^{1/2}|x|^3 + \varepsilon x^2 + \varepsilon^{3/2}|x| + \varepsilon^2\}. \quad (5.7)$$

Proof. With $t = \theta\sqrt{b_G}$ the centered phase is

$$-t^2/2 + V_3 + V_4 + V_5 + O(\varepsilon^2|t|^6),$$

where

$$V_3 = -\frac{ik_3t^3}{6b_G^{3/2}}, \quad V_4 = \frac{k_4t^4}{24b_G^2}, \quad V_5 = \frac{ik_5t^5}{120b_G^{5/2}}.$$

On $|t| \leq \varepsilon^{-1/24}$ expand the exponential through weighted degree three:

$$1 + V_3 + (V_4 + V_3^2/2) + (V_5 + V_3V_4 + V_3^3/6). \quad (5.8)$$

The omitted terms, multiplied by $e^{-t^2/2}$, have $L^1(dt)$ norm $O(\varepsilon^2)$. In detail, $|V_j| \leq C\varepsilon^{(j-2)/2}|t|^j$, the omitted monomials have weight at least four, and the sum of the retained phase magnitudes and phase remainder tends to zero uniformly on this window. Exponential Taylor's remainder is bounded by $C\varepsilon^2$ times a fixed Gaussian-integrable polynomial. The same estimate holds with any fixed extra polynomial weight in t . Lemma 2.2 handles the complementary arcs, and the Gaussian-polynomial tails permit extending the retained integrals to $\mathbb{R}$.

Fourier inversion adds e^{ixt} , of modulus one. Hence the integrated $O(\varepsilon^2)$ error is uniform in the target. The retained integrals have the following global bounds:

- The Gaussian term is $e^{-x^2/2} = 1 - x^2/2 + O(x^4)$, for every real x
- The imaginary odd term V_3 , using $\sin(xt) = xt + O(|xt|^3)$, gives $k_3\delta/(2b_G^2) + O(\varepsilon^{1/2}|x|^3)$
- The real even term $V_4 + V_3^2/2$ has value $A_1(G)$ at $x = 0$; its change is $O(\varepsilon x^2)$ by $|\cos(xt) - 1| \leq x^2t^2/2$
- The imaginary odd weight-three terms have entire integral $O(\varepsilon^{3/2}|x|)$ by $|\sin(xt)| \leq |xt|$

All the trigonometric inequalities hold on the entire real line. Thus (5.7) remains valid even at targets where the actual lattice probability is zero. $\square$

5.3 Summing the error without an exceptional-count cutoff

Under Q , compound-Poisson moments give

$$\begin{aligned} \mathbb{E}_Q \delta^2 &= v = O(r^4), & \mathbb{E}_Q |\delta| &= O(r^2), \\ \mathbb{E}_Q \delta^4 &= 3v^2 + \mathcal{D}^4 P(r) = O(r^8), & \mathbb{E}_Q |\delta|^3 &\leq (\mathbb{E}_Q \delta^4)^{3/4} = O(r^6). \end{aligned}$$

Since $b_0 := \mathcal{D}^2 C_0(r) = b - v \sim nr$, the five terms on the right of (5.7), averaged over Q , have orders

$$O(r^6/n^2), \quad O(r^5/n^2), \quad O(r^4/n^2), \quad O(r^3/n^2), \quad O(r^2/n^2), \quad (5.9)$$

respectively. This proves a summable local error over all exceptional counts; a pointwise error with an unproved cutoff is not being integrated.

For the denominator of (5.5), take $G = C$, $\delta = 0$:

$$\sqrt{2\pi b} \mathbb{P}(N = n) = 1 + A_1(C) + O(\varepsilon^2).$$

The rational formula for A_1 gives

$$A_1(C_0) - A_1(C) = O(r^4/n^2). \quad (5.10)$$

Indeed the changes in b, κ_3, κ_4 are all $O(r^4)$; their corresponding partial derivatives in A_1 are $O(n^{-2})$, $O((n^2r)^{-1})$, and $O((n^2r^2)^{-1})$. The perturbations are small fractions of the positive original quantities, so the mean value theorem applies uniformly.

Finally,

$$\begin{aligned}\sqrt{b/b_0} &= 1 + \frac{v}{2b} + O(v^2/b^2), & \frac{1}{b_0} &= \frac{1}{b} + O(v/b^2), \\ \frac{\kappa_3 - \mathcal{D}^3 P(r)}{b_0^2} &= \frac{\kappa_3}{b^2} + O(r^3/n^2 + r^4/n^2).\end{aligned}$$

The last bound is deliberately loose but sufficient after multiplication by $\mathbb{E}_Q|\delta|$. Divide the numerator expansion by the denominator and use (5.9). Every discarded product has $L^1(Q)$ size $O(r^6/n^2)$: the prefactor error is $O(v^2/b^2)$, and products with $A_1 = O(r/n)$ are smaller. This proves (5.3).

5.4 The constant and an independent tail check

Under Q , $T_3 + T_4$ is Poisson with mean $5r^4/24 \rightarrow \infty$, and the centered isolate count is negligible on the r^2 scale. Hence $\delta/\sqrt{v}$ converges to a standard normal Z . Its standardized fourth moments stay bounded, making the squares uniformly integrable. Direct Gaussian integration gives

$$\mathbb{E}_Q|1 - \delta^2/v| \rightarrow \mathbb{E}|1 - Z^2| = 4\phi(1).$$

The linear term in (5.3) has L^1 size $O(\kappa_3\sqrt{v}/b^2) = O(r^2/n) = o(v/b)$, and the remainder is also $o(v/b)$. Multiplication by the total-variation factor $1/2$ proves (5.4).

For later use, the centered cumulant generating function is

$$r(e^t - 1 - t) + \frac{5r^4}{24}(e^{4t} - 1 - 4t) \leq Cr^4t^2$$

for $|t|$ below a fixed constant. Chernoff's bound with t a sufficiently small constant times $1/r$ yields

$$Q(|\delta| > r^3) \leq 2e^{-cr^2}. \quad (5.11)$$

Also $h \leq 1/\mathbb{P}(N = n) = O(\sqrt{b})$, so the true exceptional marginal has tail $e^{-cr^2+O(r)}$, smaller than every fixed power of n^{-1} . This tail argument is useful later, but was not needed to justify the summation in (5.9).

6 Rare deletion probabilities and their saddle shift

For a fixed nonzero polynomial P with nonnegative coefficients, of degree $d \geq 1$, suppose $C - P$ still has nonnegative coefficients. Here P is used afresh and need not be the full exceptional polynomial of Section 5. When P records selected component categories, the ratio $[z^n]e^{C-P}/[z^n]e^C$ is their absence probability. The analytic formula below also applies to any polynomial satisfying the stated coefficient conditions. Set $p_j = \mathcal{D}^j P(r)$.

Theorem 6.1 (Fixed-polynomial deletion). *At the original exact saddle r ,*

$$\log \frac{[z^n]e^{C(z)-P(z)}}{[z^n]e^{C(z)}} = -P(r) + \frac{p_2 - p_1^2}{2b} - \frac{p_1\kappa_3}{2b^2} + O_P(r^{3d-1}/n^2). \quad (6.1)$$

Proof. Let ρ solve $\mathcal{D}(C - P)(\rho) = n$ and write $s = \log(\rho/r)$. Since deleting a nonzero nonnegative polynomial lowers the mean at r , monotonicity gives $s > 0$. On $0 \leq s \leq 2p_1/b$, cumulant estimates

stay uniform because $rs = O(r^d/n) = o(1)$. The increase in the mean is $bs(1 + o(1))$, so this interval brackets the new saddle. Taylor expansion then improves the estimate to

$$s = \frac{p_1}{b} + O(r^{2d-1}/n^2), \quad s = O(r^{d-1}/n). \quad (6.2)$$

The saddle phase changes by

$$\begin{aligned} (C - P)(\rho) - n \log \rho - C(r) + n \log r \\ = -P(r) - \frac{p_1^2}{2b} + O(r^{3d-1}/n^2). \end{aligned} \quad (6.3)$$

For clarity, the quadratic combination is $-p_1s + bs^2/2$, and the cubic and perturbation remainders are bounded by κ_3s^3 and p_2s^2 . Their orders are r^{3d-1}/n^2 and r^{3d-2}/n^2 .

The new variance is

$$\mathcal{D}^2(C - P)(\rho) = b - p_2 + \frac{\kappa_3 p_1}{b} + O(r^{2d+1}/n).$$

Taking the logarithm of the square-root variance ratio gives

$$\frac{p_2}{2b} - \frac{\kappa_3 p_1}{2b^2} + O(r^{2d}/n^2). \quad (6.4)$$

The apparently larger variance remainder is divided by $b \asymp nr$ in this step.

The first saddle correction changes by $O(r^{d+1}/n^2)$ under the shift and the direct polynomial deletion. One can check it by differentiating (3.3):

$$\mathcal{D}E_1 = \frac{\kappa_5}{8b^2} - \frac{2\kappa_3\kappa_4}{3b^3} + \frac{5\kappa_3^3}{8b^4} = O(r^2/n).$$

Multiply by (6.2); direct deletion is smaller. Applying Theorem 3.1 through $J = 1$ leaves $O(r^2/n^2)$ in the logarithm. For $d \geq 1$, all these errors fit inside $O(r^{3d-1}/n^2)$. Combining (6.3) and (6.4) proves the result. $\square$

6.1 No nonsunflower components

Take $P(z) = 5z^4/24$. Then

$$\log \mathbb{P}_n(T_3 = T_4 = 0) = -\frac{5r^4}{24} - \frac{25r^8}{72b} + \frac{5r^4}{3b} - \frac{5r^4\kappa_3}{12b^2} + O(r^{11}/n^2). \quad (6.5)$$

The leading saddle-shift correction is $-25r^7/(72n)(1 + O(1/r))$. Exponentiation gives the relative estimate

$$\mathbb{P}_n(T_3 = T_4 = 0) = e^{-5r^4/24} \left\{ 1 - \frac{25r^8}{72b} + O(r^4/n + r^{14}/n^2) \right\}. \quad (6.6)$$

The r^{14}/n^2 term includes the square of the leading logarithmic correction. This relative result does not follow from the absolute total-variation estimate.

6.2 No isolated vertices

Taking $P(z) = z$ gives the exact event $S = 0$ and

$$\log \frac{B_n}{A_n} = -r + \frac{r - r^2}{2b} - \frac{r\kappa_3}{2b^2} + O(r^2/n^2). \quad (6.7)$$

In particular $B_n/A_n \sim e^{-r} \rightarrow 0$. The no-isolate restriction retains a vanishing fraction of all admissible objects and removes almost all of them. This is compatible with isolates making up a vanishing fraction of the components: absence of a growing but sparse category can still be a very rare event.

7 Projected Gaussian laws on two different scales

The raw edge/component pair is asymptotically singular. The defect (1.4) identifies the residual direction and makes a nonsingular limit available before taking any Gaussian approximation.

Set

$$d = r + \frac{r^4}{4}, \quad \tau^2 = r + \frac{r^4}{3}, \quad \eta = r + r^4, \quad (7.1)$$

$$V = C(r) - \frac{n^2}{b}, \quad T = \tau^2 - \frac{\eta^2}{b}, \quad G_* = d - \frac{n\eta}{b}, \quad \rho = \frac{G_*}{\sqrt{VT}}. \quad (7.2)$$

These are Boltzmann means and covariance Schur complements for conditioning on total vertex count. Their scales are

$$V \sim \frac{n}{r^2}, \quad T \sim \frac{r^4}{3}, \quad \rho \sim \frac{\sqrt{3}}{4} \frac{r^3}{\sqrt{n}} \rightarrow 0. \quad (7.3)$$

For the cancellation in V , the H -only expression is exactly

$$H - \frac{(\mathcal{D}H)^2}{\mathcal{D}^2H} = \frac{Hr}{r^2 + 5r + 4},$$

and the polynomial changes contribute $O(r^4)$. The other equivalents follow directly from (2.1).

Theorem 7.1 (Quantitative projected characteristic law). *For every fixed $T_0 < \infty$, uniformly over $|t|, |s| \leq T_0$,*

$$\begin{aligned} \mathbb{E}_n \exp \left\{ it \frac{K - C(r)}{\sqrt{V}} + is \frac{\Delta - d}{\sqrt{T}} \right\} \\ = \exp \left\{ -\frac{1}{2}(t^2 + s^2 + 2\rho ts) \right\} + O_{T_0}(r^{-2} + \sqrt{r/n}). \end{aligned} \quad (7.4)$$

Thus the two standardized variables converge jointly to independent standard normals. In particular,

$$\left(\frac{K - C(r)}{\sqrt{V}}, \frac{E - [n - 2C(r) + d]}{2\sqrt{V}} \right) \Rightarrow (Z, -Z), \quad (7.5)$$

whereas the residual $E + 2K - n$ fluctuates independently on scale $r^2/\sqrt{3}$.

7.1 Orthogonal projection before conditioning

Index the connected component types by α , with Boltzmann mean λ_α , size m_α , and defect d_α . The latter is one for isolates and three-edge tetrahedra, two for complete tetrahedra, and zero otherwise. Their counts Z_α are independent Poissons. Direct sums give

$$\begin{aligned} \text{Var } K &= C(r), & \text{Cov}(K, N) &= n, & \text{Var } N &= b, \\ \text{Var } \Delta &= \tau^2, & \text{Cov}(\Delta, N) &= \eta, & \text{Cov}(K, \Delta) &= d. \end{aligned}$$

Define the projection coefficients

$$a_\alpha = 1 - \frac{n}{b}m_\alpha, \quad q_\alpha = d_\alpha - \frac{\eta}{b}m_\alpha,$$

and the centered sums

$$Y = \sum_{\alpha} a_\alpha (Z_\alpha - \lambda_\alpha), \quad Q_* = \sum_{\alpha} q_\alpha (Z_\alpha - \lambda_\alpha).$$

Both have zero covariance with N , and

$$\text{Var } Y = V, \quad \text{Var } Q_* = T, \quad \text{Cov}(Y, Q_*) = G_*.$$

On $N = n$, they equal $K - C(r)$ and $\Delta - d$. The conditional characteristic numerator is therefore the Fourier integral of

$$\exp \sum_{\alpha} \lambda_{\alpha} \{e^{iu_{\alpha}} - 1 - iu_{\alpha}\}, \quad u_{\alpha} = \frac{xm_{\alpha}}{\sqrt{b}} + \frac{ta_{\alpha}}{\sqrt{V}} + \frac{sq_{\alpha}}{\sqrt{T}}, \quad (7.6)$$

where $x = \theta\sqrt{b}$. Its quadratic exponent is exactly

$$-\frac{1}{2}(x^2 + t^2 + s^2 + 2\theta ts). \quad (7.7)$$

The absence of xt and xs terms is the reason for projecting first.

7.2 Cubic error estimates

The absolute third-moment sums satisfy

$$\begin{aligned} \sum_{\alpha} \lambda_{\alpha} (m_{\alpha}/\sqrt{b})^3 &= O(\sqrt{r/n}), \\ \sum_{\alpha} \lambda_{\alpha} |a_{\alpha}/\sqrt{V}|^3 &= O(\sqrt{r/n}), \\ \sum_{\alpha} \lambda_{\alpha} |q_{\alpha}/\sqrt{T}|^3 &= O(r^{-2}). \end{aligned} \quad (7.8)$$

The first is immediate from $\kappa_3/b^{3/2}$. For the second, the main H weights have $m = 2 + \text{Pois}(r)$ and

$$\frac{n}{b} = \frac{r+2}{r^2+5r+4} + \text{an exponentially small polynomial error.}$$

Consequently $\mathbb{E}|1 - (n/b)m|^3 = O(r^{-3/2})$. One elementary bound is $\mathbb{E}|\text{Pois}(r) - r|^3 \leq (3r^2 + r)^{3/4}$, from the fourth moment. Multiply by $H \asymp n/r$ and divide by $V^{3/2} \asymp n^{3/2}/r^3$. The finitely modified

small sizes contribute at most $O(r^7/n^{3/2})$, a smaller order. For the third estimate, the exceptional types contribute $O(r^4)/T^{3/2} = O(r^{-2})$; the ordinary types contribute at most

$$\frac{(\eta/b)^3 \kappa_3}{T^{3/2}} = O(r^5/n^2).$$

The all-real Taylor inequality $|e^{iu} - 1 - iu + u^2/2| \leq |u|^3/6$, together with $(|u| + |v| + |w|)^3 \leq 9(|u|^3 + |v|^3 + |w|^3)$, bounds the phase difference between (7.6) and (7.7) by

$$C(|x| + |t| + |s|)^3 \{r^{-2} + \sqrt{r/n}\}. \quad (7.9)$$

7.3 Domination on the full marked contour

The estimates must remain uniform for bounded t, s outside a tiny Gaussian window. On $|\theta| \leq c_0/r$, retain ordinary sizes $m \in [r/2, 2r]$ and all exceptional types. Their phases in (7.6) are at most one in absolute value for sufficiently small c_0 and large n . The normalized covariance matrix of these retained types differs from the full matrix by $o(1)$ in operator norm. Indeed the omitted Poisson size tails are exponentially small in r , even after the polynomial factors from normalizing the small variance V ; and essentially all defect variance is retained. Since $\rho \rightarrow 0$, its quadratic form is bounded below by a positive constant times $x^2 + t^2 + s^2$. Using $1 - \cos u \geq cu^2$ yields domination by e^{-cx^2} .

For the outer arc, ordinary component phases take the form $m\theta' + u$, where

$$\theta' = \theta - \frac{tn}{b\sqrt{V}} - \frac{s\eta}{b\sqrt{T}}, \quad u = \frac{t}{\sqrt{V}}.$$

The shift is $O(n^{-1/2} + r/n) = o(1/r)$. For $c_0/r \leq |\theta| \leq \pi$, the distance of θ' from zero modulo 2π remains at least a constant times $1/r$. The radial estimate (2.5) applies unchanged to $e^{iu}H(re^{i\theta'})$, since its modulus is unchanged. The exceptional and other polynomial terms add only $O(r^4)$. Thus this entire outer arc is $O(e^{-cn/r^2})$ uniformly for bounded t, s .

7.4 Completion of the Gaussian proof

For complex z, w ,

$$|e^z - e^w| \leq |z - w|e^{\max(\Re z, \Re w)}$$

by integrating along the line segment. Apply this inequality to the two exponents, use (7.9) and the preceding domination, and integrate against e^{-cx^2} . The numerator error is the normalized integral error $O_{T_0}(r^{-2} + \sqrt{r/n})$ times $(2\pi b)^{-1/2}$. The denominator is

$$\mathbb{P}(N = n) = (2\pi b)^{-1/2}(1 + O(r/n)).$$

The Gaussian integral of (7.7) therefore gives (7.4). Finally $\sqrt{T}/\sqrt{V} = O(r^3/\sqrt{n}) \rightarrow 0$, so (1.4) yields (7.5) and completes the proof of Theorem 7.1.

8 Hybrid independence and moment refinements

A joint Gaussian limit for normalized exceptional counts would lose their discrete structure. The next statement keeps the entire growing-mean Poisson vector X and tests it against every bounded function.

Theorem 8.1 (Mixed Gaussian–Poisson test bound). *For every fixed T_0 and every complex function g on $\mathbb{N}_0^3$ with $|g| \leq 1$,*

$$\sup_{|t| \leq T_0} \left| \mathbb{E}_n \left[e^{it(K-C(r))/\sqrt{V}} g(X) \right] - e^{-t^2/2} \mathbb{E}_Q g(X) \right| = O_{T_0}(r^3/\sqrt{n}), \quad (8.1)$$

with a constant uniform over g .

This is a mixed characteristic-function/total-variation test class. It does not assert total-variation convergence of a discrete component count to a continuous Gaussian, which would be false.

Proof. On $|W - \mu| \leq r^3$, condition on X . The remaining assembly has EGF e^{C_0} and size $n - W$; let r_W be its own saddle. The monotonic bracketing and Taylor argument used in (6.2), now allowing either sign, gives uniformly

$$\log(r_W/r) = -\frac{W - \mu}{b_0} + O\left(\frac{r(W - \mu)^2}{b_0^2}\right), \quad b_0 = \mathcal{D}^2 C_0(r). \quad (8.2)$$

In particular $r_W/r \rightarrow 1$ and $n - W \sim n$ uniformly. The one-dimensional projected Fourier proof from Section 7, with q_α omitted, applies to this remaining assembly with uniform constants. Its component count K_0 has a Gaussian characteristic approximation centered at $C_0(r_W)$, with variance

$$V_W = C_0(r_W) - \frac{[\mathcal{D}C_0(r_W)]^2}{\mathcal{D}^2 C_0(r_W)}$$

and error $O_{T_0}(\sqrt{r/n})$.

Write $L_X = S + T_3 + T_4$ and $\Lambda = \mathbb{E}_Q L_X = r + 5r^4/24$. Taylor expansion using (8.2) gives

$$\begin{aligned} L_X + C_0(r_W) - C(r) &= L_X - \Lambda - \frac{n - \mu}{b_0}(W - \mu) \\ &\quad + O\left(\frac{(W - \mu)^2}{nr}\right), \end{aligned} \quad (8.3)$$

and uniformly on this window,

$$\frac{V_W}{V} = 1 + O(r^6/n). \quad (8.4)$$

For (8.4), deleting the fixed polynomial changes the Schur variance by $O(r^4)$, while $\mathcal{D}V = O(n/r)$ and the saddle shift changes it by $O(|W - \mu|/r^2)$. Dividing by $V \asymp n/r^2$ proves the displayed bound.

Under Q the expected absolute value of the linear part of (8.3) is $O(r^2)$: both centered variables have standard deviations $O(r^2)$ and $(n - \mu)/b_0 = O(1/r)$. Its quadratic remainder has expectation $O(r^3/n)$. After division by $\sqrt{V}$, the expected centering change is $O(r^3/\sqrt{n})$. The inequality $|e^{iu} - 1| \leq |u|$ and (8.4) therefore replace the conditional Gaussian by $e^{-t^2/2}$ with this error after averaging. Its changed Fourier parameter is $t\sqrt{V_W/V}$, still in a fixed bounded interval.

To justify the averaging measure, first replace the true marginal of X by Q while the exact conditional integrand remains bounded by one. Theorem 5.1 makes this cost $O(r^3/n)$, uniformly over all $|g| \leq 1$. Then estimate the centering changes under Q as above. Both marginals' probabilities outside the window are $e^{-cr^2+O(r)}$ by (5.11). Finally $r^6/n = o(r^3/\sqrt{n})$, so all errors fit in (8.1). $\square$

8.1 Fixed-target cancellation and defect moments

For a fixed nonnegative integer m define $R_m = \mathbb{P}(N = n - m)/\mathbb{P}(N = n)$, where N is the unconditioned Boltzmann total at r . A finer cancellation than the global local bound gives

$$R_m = 1 + \frac{m\kappa_3}{2b^2} - \frac{m^2}{2b} + O_m(r/n^2). \quad (8.5)$$

Here one must not subtract unrelated $O(\varepsilon^2)$ errors. Subtract the Fourier integrals at shifts m and zero before estimating. Their common central remainder from (5.8) has $L^1(|t|dt)$ norm $O(\varepsilon^2)$, by the same Gaussian-polynomial bound used in Lemma 5.2. Multiplication by $e^{imt/\sqrt{b}} - 1$ gives an extra factor $m/\sqrt{b}$, so this contribution is $O(\varepsilon^2 m/\sqrt{b}) = o(r/n^2)$. The retained odd weight-three terms cost $O(\varepsilon^{3/2} m/\sqrt{b}) = O(r/n^2)$, and the change of even weight-two terms costs $O(\varepsilon m^2/b) = O(n^{-2})$. Gaussian Taylor terms and denominator division are no larger. The full-arc tails remain negligible. This proves (8.5) with its improved remainder.

Corollary 8.2 (Conditioning corrections to defect moments). *At the original saddle,*

$$\mathbb{E}_n \Delta = d + \frac{\eta\kappa_3}{2b^2} - \frac{r + 4r^4}{2b} + O(r^5/n^2), \quad (8.6)$$

$$\text{Var}_n \Delta = \tau^2 - \frac{\eta^2}{b} + O(r^4/n + r^9/n^2) = T + O(r^4/n). \quad (8.7)$$

Proof. For each exceptional type α , the Poisson factorial-moment identity gives

$$\mathbb{E}_n Z_\alpha = \lambda_\alpha R_{m_\alpha}, \quad \mathbb{E}_n (Z_\alpha Z_\beta - \mathbf{1}_{\alpha=\beta} Z_\alpha) = \lambda_\alpha \lambda_\beta R_{m_\alpha + m_\beta}.$$

All sizes here are one or four, so (8.5) is uniform over the finitely many required targets. The weighted sums are

$$\sum d_\alpha \lambda_\alpha = d, \quad \sum d_\alpha m_\alpha \lambda_\alpha = \eta, \quad \sum d_\alpha m_\alpha^2 \lambda_\alpha = r + 4r^4.$$

These give (8.6). For the variance, writing the denominator variance as b ,

$$R_{a+c} - R_a R_c = -\frac{ac}{b} + O(r/n^2)$$

for fixed a, c . Summing weighted pairs gives $-\eta^2/b + O(r^9/n^2)$; the diagonal contribution is $\tau^2 + O(r^4/n)$. Since $r^5/n \rightarrow 0$, the last simplification in (8.7) follows. $\square$

The use of Boltzmann centers and Schur variances in Theorem 7.1 is therefore consistent with the actual conditional moments at the stated scales. Also $K \sim n/r$ in probability, while $S + T_3 + T_4 = O_{\mathbb{P}_n}(r^4)$ by Theorem 5.1. Thus exceptional components form a vanishing fraction of all components in probability. No coupling across different n , and no almost-sure assertion, is being made.

9 Transfer to the no-isolate model

Let $C_B = C - z$, and let r_B solve $\mathcal{D}C_B(r_B) = n$. Every result has a no-isolate version at this saddle. The distinction between using the same leading scale and using the same saddle is essential: (6.7) shows $B_n/A_n \sim e^{-r}$, far from one.

Theorem 9.1 (The A323296 versions). *For the uniform model counted by B_n , replace C by C_B , r by r_B , and κ_j, b by the corresponding cumulants. Theorem 3.1 and the inverse theorems remain valid. Delete S from the exceptional vector and use*

$$\begin{aligned} \lambda_3 &= r_B^4/6, & \lambda_4 &= r_B^4/24, & W &= 4T_3 + 4T_4, \\ \mu &= 5r_B^4/6, & v &= 10r_B^4/3, \\ d &= r_B^4/4, & \tau^2 &= r_B^4/3, & \eta &= r_B^4. \end{aligned}$$

The size-squared defect sum also loses its isolate term:

$$\sum_{\alpha} d_{\alpha} m_{\alpha}^2 \lambda_{\alpha} = 4r_B^4, \quad \mathbb{E}_{B,n} \Delta = d + \frac{\eta \kappa_3}{2b^2} - \frac{4r_B^4}{2b} + O(r_B^5/n^2).$$

With these replacements, the likelihood, sharp total-variation, deletion, projected Gaussian, hybrid independence, and fixed-defect moment conclusions hold. In particular,

$$d_{\text{TV}}\left(\text{Law}_{B,n}(T_3, T_4), \text{Pois}(r_B^4/6) \otimes \text{Pois}(r_B^4/24)\right) \sim \frac{10}{3} \phi(1) \frac{r_B^3}{n},$$

and the exact residual is $E + 2K - n = T_3 + 2T_4$. Its Gaussian scale is $r_B^2/\sqrt{3}$, independently of the component-count scale $\sqrt{n}/r_B$.

Proof. The connected EGF is still H plus a fixed polynomial with nonnegative coefficients and all sufficiently large consecutive sizes present. Thus the cumulant and full-arc proofs apply verbatim. The exceptional polynomial becomes $5z^4/24$; its compound-Poisson moments have the same leading orders and the same dominant variance $10r_B^4/3$. Every step of the summable likelihood argument and its sharp constant is unchanged. The defect, covariance, projection, and hybrid proofs use only the displayed component sums and the same H -dominated ordinary types, so the replacements give the claimed statements.

Removing z affects the reduced inverse calculation only by exponentially small terms in r , below every fixed inverse-logarithmic order. Therefore the coefficients in (4.8) and its recursion are unchanged. For the fine inverse one must instead use C_B in (4.1). Finally the coefficient formula at adjacent integers gives

$$\frac{B_{n+1}}{B_n} \sim \frac{n}{r_{B,n}} \longrightarrow \infty,$$

so the sequence is eventually strictly increasing and the two-ceiling proof applies beyond its finite initial segment. $\square$

10 Exact checks and reproducible computation

The archive separates finite exact statements from noncertified numerical observations. Neither kind of finite check establishes an asymptotic remainder or replaces the proofs above. The accompanying `COMPUTATION.md` records bounds, algorithms, and environment requirements in detail.

10.1 Exact coefficient and component checks

The labelled assembly recurrence is

$$A_0 = 1, \quad A_n = \sum_{m=1}^n \binom{n-1}{m-1} c_m A_{n-m}. \quad (10.1)$$

It follows by choosing the component containing vertex one. Setting $c_1 = 0$ gives B_n . An independent coefficient route expands

$$e^{C(z)} = e^{U(z)} \sum_{k \geq 0} \frac{z^{2k} e^{kz}}{2^k k!},$$

where only $k \leq n/2$ contributes to $[z^n]$. A short polynomial-exponential recurrence computes e^U , and exact rational or integer convolutions recover the labelled coefficients. Agreement of these two routes is stronger than checking two rearrangements of (10.1).

The initial terms are

n	0	1	2	3	4	5	6	7	8
A_n	1	1	1	2	16	76	271	1212	10158
B_n	1	0	0	1	11	10	25	406	4823

Direct enumeration of admissible edge sets through $n = 8$ independently checks these terms, the connected multiplicities, and (1.4) on every object. Exact marked first and second moment identities are checked by two coefficient routes. The released bound for the recurrence-based count and moment checks is $n = 640$; this is a deliberate resource bound, not a limitation of the mathematical formulas.

10.2 Symbolic and numerical checks

Exact rational Gaussian contractions in (3.1) reproduce (3.3)–(3.4). Independent formal series reversal reproduces all coefficients in (4.11) and (4.8). The public symbolic program reports zero differences, with bounded expansion orders and explicit validation.

The separately labelled `diagnostics.py` evaluates exact-count ratios and saddle approximations at finite sizes. Its tables include the scaled total-variation ratio, the L^1 likelihood residual, deletion corrections, coefficient residuals, and marked covariance scales. These are floating-point calculations, explicitly *noncertified*: no interval enclosure, rigorous numerical error bar, or finite onset of the asymptotic estimates is claimed. Exact large integers are not converted to unbounded decimal strings merely to make these diagnostics.

For a compact numerical illustration, let $c_* = (10/3)\phi(1)$ and let R be the likelihood remainder in (5.3). The following rounded values are taken from the *noncertified* diagnostic receipt; Poisson sums are truncated as described in the supplement.

n	$d_{TV}/(\phi(1)v/b)$	$d_{TV}/(c_*r^3/n)$	$n^2\mathbb{E}_Q R /r^6$	$\text{Var}_n(K)/V$
80	1.14453	0.52610	0.97029	1.00009
160	1.11414	0.54980	1.08081	0.99947
320	1.08574	0.57171	1.19548	0.99961
640	1.06423	0.59317	1.30935	0.99984

The two TV normalizations are asymptotically equivalent but differ substantially at these sizes. The table illustrates why exact-saddle quantities can be useful well before their leading logarithmic simplifications are accurate; it certifies neither a convergence rate nor an asymptotic onset.

[write] **The table recomputed, 7 October 2026.** The write recomputed all sixteen entries with its own code: the law of X from the exact formula (5.5) with exact coefficients of e^{C_0} and e^C , summed over

the Poisson product (terms of Q -mass below 10^{-35} dropped), and $\text{Var}_n(K)$ from exact first and second factorial moments. Every printed value is the correct rounding of the recomputed one, which also equals `data/diagnostic_receipt.json` to the printed digits. The printed values that are not truncations are 1.08081 and 0.99947 ($n = 160$; truncations 1.08080, 0.99946), 0.57171 ($n = 320$; 0.57170), 1.30935 and 0.99984 ($n = 640$; 1.30934, 0.99983).

10.3 Reproduction boundaries

The build verifies the complete source inventory before running checks, compiles a disposable source copy with a private TeX format and shell escape disabled, and requires byte-for-byte agreement with the frozen PDF and receipts. Python child processes disable bytecode output and enforce a decimal conversion cap independently of the inherited environment. Public input bounds use explicit exceptions, so optimization mode does not remove validation.

The ZIP replay extracts the actual archive, verifies its contents and metadata, rebuilds it in normal and optimized Python modes, and compares the PDF, receipts, and all regenerated archive members. Source hashes are checked before and after builds. Reproducibility is relative to the recorded Python, symbolic-library, and TeX toolchain; a different engine or font installation may legitimately produce different PDF bytes and will be reported as a mismatch. The package includes no third-party paper PDFs and no private working or review notes.

[write] **The shipped files, 7 October 2026.** In ProveIt the PDF, the manifest and the delivered README are not shipped; the programs are `code/*.py` (the builder as `code/build.py`), and the four receipts and the requirements are shipped under `data/`. The checkers write to standard output and run from the shipped files; the builder and the archive replay expect the delivered layout and the manifest, so they run only in a re-extracted archive. The report README gives commands.

Remark 10.1 ([write] The OEIS entries, 7 October 2026). The write read the entries in the internal format on 7 October 2026. *A323297*, revision #14 (18 August 2019), by Gus Wiseman (11 January 2019): “Number of 3-uniform hypergraphs on n labeled vertices where no two edges have exactly one vertex in common.”, with the e.g.f. by Andrew Howroyd (18 August 2019), “Binomial transform of A323296.”, and a b-file by Howroyd for $n = 0, \dots, 200$. *A323296*, revision #15 (18 August 2019), by Wiseman (11 January 2019), the spanning version, with Howroyd’s comment of 18 August 2019 that such a hypergraph is a disjoint union of components in which any two edges share two vertices (A323294), his e.g.f. and a b-file for $n = 0, \dots, 200$. The component decomposition is thus stated in A323296; A323297 carries the e.g.f. *A323294*, revision #34 (14 November 2025), by Wiseman (10 January 2019), with $a(n) = \binom{n}{2}$ for $n \geq 5$ by Wiseman (16 January 2019), the e.g.f. by Howroyd (2019) and a b-file for $n = 0, \dots, 1000$. None of the three entries has an asymptotic formula or a conjecture. *Checks.* All b-file terms equal the write’s computation. [Independent check, 7 October 2026: for A323294 this holds with its spanning conventions $a(0) = 1$, $a(1) = 0$; the report’s $c_1 = 1$ counts the isolated vertex.] No OEIS edit is part of this work.

11 Further questions

The results isolate several extensions with distinct analytic obstacles.

Effective constants and certified integer inversion. The two-ceiling theorem becomes a usable certified large-range threshold procedure once the full-arc and local Taylor constants are explicit beyond a concrete saddle. It would be useful to combine interval evaluation of F_J with a

provable tail integral bound and exact adjacent recurrence checks. Finite numerical agreement does not provide those constants. An implementation should also control cancellation when evaluating the small difference $V = C - n^2/b$.

Higher conditional likelihood corrections. The local Fourier proof can retain higher weighted degrees, yielding polynomial corrections in the centered removed mass. The main additional issue is to organize these corrections so that their signed Q -mass cancels order by order and their global remainder stays summable under a growing-parameter compound Poisson law. Such an expansion could refine the sharp total-variation equivalent beyond its first constant. It would also be natural to turn a corrected signed approximation into a genuine probability law without losing the claimed order.

An optimal hybrid error. The bound $O(r^3/\sqrt{n})$ in Theorem 8.1 is driven by the exceptional-count contribution to the conditional component mean. One may ask whether explicitly centering K by that observed contribution gives a smaller uniform mixed-test error. A result in a stronger metric would need to respect the lattice nature of K rather than compare it in total variation directly to a continuous law.

Moderate and large edge deviations. The two Gaussian scales describe bounded normalized fluctuations. Tilting the defect marker may change the balance between the rare tetrahedral components and the ordinary pair-stars. A uniform marked saddle analysis with growing Fourier or exponential parameters could locate the crossover between Gaussian fluctuations, Poisson moderate deviations, and genuine large deviations. The compact-parameter proof here does not imply such a result.

Larger uniformity and structural obstructions. For 3-uniform systems the component classification collapses to pair-stars and two finite exceptions. For higher uniformity, forbidden singleton intersections admit more complicated connected families; the simple polynomial-times-exponential connected EGF may disappear. Determining which restricted connected subclasses still admit a comparable explicit Poisson correction would connect enumeration more directly with the structural questions in the extremal literature.

Growing deletions. Theorem 6.1 fixes the deleted polynomial independently of n . Allowing its degree or coefficients to grow would test the boundary of the small saddle displacement p_1/b . A uniform theorem would need both a quantitative nonnegative-coefficient condition and control of how the deleted mass changes every Fourier arc. It cannot be obtained merely by replacing the fixed degree in the displayed big- O bound by a function of n .

[write] **Added 7 October 2026, under Vladimir's standing rule of 4 October 2026.** These six directions are the source's own and stay open. The write adds from the non-claims explicit constants for Theorem 4.1 and certified diagnostics. No claim of the source was found false, and the OEIS entries contain no conjecture.

References

- [1] The OEIS Foundation Inc., *A323297: Number of 3-uniform hypergraphs on labelled vertices where no two edges have exactly one vertex in common*, <https://oeis.org/A323297>. Exponential generating function credited to Andrew Howroyd, 18 August 2019. Checked 5 October 2026.
- [2] The OEIS Foundation Inc., *A323296: The spanning no-singleton-intersection 3-uniform hypergraph count*, <https://oeis.org/A323296>. Component interpretation and exponential generating function credited to Andrew Howroyd, 18 August 2019. Checked 5 October 2026.
- [3] The OEIS Foundation Inc., *A323294: Spanning triple systems with pairwise intersection of size two*, <https://oeis.org/A323294>. The eventual formula $\binom{n}{2}$ is credited to Gus Wiseman, 16 January 2019; the related EGF to Andrew Howroyd, 18 August 2019. Official sequence data inspected in the source review.
- [4] A. M. Odlyzko and L. B. Richmond, *Asymptotic expansions for the coefficients of analytic generating functions*, *Aequationes Mathematicae* **28** (1985), 50–63. doi:10.1007/BF02189391. Author copy: <https://www-users.cse.umn.edu/~odlyzko/doc/arch/admissible.fns.pdf>. Theorems 3–4 give the Harris–Schoenfeld complete-expansion framework and its exponential closure. The direct proofs in this article provide the explicit specialization and stated error scales.
- [5] R. Arratia and S. Tavaré, *Independent process approximations for random combinatorial structures*, *Advances in Mathematics* **104** (1994), 90–154. doi:10.1006/aima.1994.1022. <https://arxiv.org/abs/1308.3279>; author journal copy: <https://www.columbia.edu/cu/simontavare/STpapers-pdf/AT94.pdf>. Sections 2–3 and 6 contain conditioned independent processes, component-type refinement, and exact total-variation reductions. The arXiv HTML’s placeholder volume and page line is not the journal citation.
- [6] P. Keevash, D. Mubayi, and R. M. Wilson, *Set systems with no singleton intersection*, *SIAM Journal on Discrete Mathematics* **20** (2006), 1031–1041. doi:10.1137/050647372. <https://people.maths.ox.ac.uk/keevash/papers/no-singleton-journal.pdf>. An extremal size and stability study, included to distinguish that question from the labelled total count considered here.
- [7] ProveIt, *Transseries and inversion*, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`; Part 0 (`p0:def:model`, `p0:def:three-inverses`, `p0:thm:staircase`) and the chapter on the Lambert template (`plt:def:lw-monomial-log-datum`, `plt:thm:lw-template`). [Added in the write.]